

1. Executive verdict

Verified fact. I reviewed the attached `multi-catfish-c2-chatgpt-review-package-20260902-r1` as a read-only package, following `START-HERE.md`, `EVIDENCE-MAP.md`, and `SOURCE-PATH-MAP.md`. I did not open the TEST split, run the simulator, train or replay a learner, or modify package files. The archive's own 109-entry manifest passed in full.

`multi-catfish-c2-chatgpt-review...`

Verified fact. The two completed experiments constitute a valid **six-physical-world, three-lineage, TRAIN-only direction screen for the two oracle-Q2 formulas that were actually run**. Recomputing pooled raw bits and raw energy from the packaged episode rows reproduces all six reported directions, including the lineage signs. The result does not authenticate learned efficacy, held-out generalization, or the unopened current exact OPS-3 implementation. See `INTEGRATION-VERIFICATION.md:30-54`, `fable-cleanroom-lane/oracle/stage1b-report.md:159-203`, and `START-HERE.md:56-93`.

Verified fact. C2 shows repeatable oracle headroom in both completed screens:

Label	Completed C2 object	FULL versus DROP-C2 EE	Lineages	Physical worlds	C2 service contrast
Verified fact	H-A deterministic hold-horizon surplus	+11.699953%	3/3 positive	6/6 positive	-1 served user-step out of 18,000
Verified fact	Fable lane's OPS-3 reading	+9.479997%	3/3 positive	6/6 positive	+1 served user-step

Verified fact. Neither completed route passes the preferred three-head method gate. Frozen C3 is negative in 0/3 lineages under both new-Q2 contexts, both routes receive `C3_CONTEXT_FAIL`, and both overall service guards fail. The current exact OPS-3 formula has mechanics evidence only; its declared outcome worlds remain unopened. See `START-HERE.md:74-93` and `fable-cleanroom-lane/oracle/stage1b-report.md:181-229`.

Inference. The evidence does **not** support stopping C2 structurally. Two materially different oracle-Q2 constructions produce positive C2 direction in every tested physical world and lineage. It does, however, support treating the current C3 as the immediate method-level blocker: C3 remains harmful when the old harmful Q2 is removed, and its isolated fresh-world contribution is approximately -9.716%, reproducing the sealed -10.286% direction.

Proposal. Hold C2 at the formula/oracle level, make the pre-outcome $h=0$ persistence decision described below, and redesign C3 before any C2 learner gate. Preserve the three-head acceptance structure. Do not accept a two-head method, weaken the three-marginal criterion, or stop C2 based on this development evidence.

2. Evidence and method audit

2.1 Authentication and source closure

Verified fact. The packaged integration records the expected H-A contract SHA-256

```
2ac39ce557ed23596728d9a83619ce585c9c30e308559bd20f48d08c0e6d1545
```

and Fable OPS-3 addendum SHA-256

```
b538932006f7e597ee199fb0701a4473587afa81b4580ca76e1dff13f9ea7046
```

The packaged runner also matches the runner hash recorded in both result JSONs. See `INTEGRATION-VERIFICATION.md:12-21` and the metadata blocks in the two result JSONs.

Verified fact. The integration record states that all 25 entries in the original Fable receipt manifest were independently checked. The reduced review package does not contain all 25 payloads at their original receipt locations: raw census NPZ material, duplicate row JSONLs, and several execution products were intentionally not included. Both result JSONs nevertheless retain the raw per-episode rows needed to recompute all reported EE and service directions. See `SOURCE-PATH-MAP.md:37-59`.

Inference. This is a **source-closure limitation**, not an arithmetic invalidation. It prevents the review package from serving as a complete binary-level reproduction of the run and prevents absolute exclusion of every possible omitted support-module difference. It does not negate the internal consistency of the captured rows, contracts, runner hash, world receipts, and ratio-of-sums recomputation.

Verified fact. Strict source closure also failed because the run's recorded source state is not identical to the earlier sealed V0.4 source snapshot. The package explicitly limits the evidence to development-direction status rather than presenting the run as a sealed confirmatory result. See `INTEGRATION-VERIFICATION.md:23-26` and `PACKAGE-MANIFEST.md:22-47`.

2.2 What the completed screen validly establishes

Label	Audit item	Adjudication
Verified fact	Endpoint arithmetic	Arm EE is computed as total delivered bits divided by total network energy after separately summing raw numerators and

Label	Audit item	Adjudication
		denominators. It is not an average of episode EEs. Per-step rows reproduce the arm totals using the canonical <code>47 × 0.640 s</code> timing.
Verified fact	Common worlds	All arms within a block use the same six evaluation seeds, starting epoch, initial state, action masks, keyed fading identities, and physical component receipts.
Verified fact	Three lineages	Each world is evaluated against three frozen head lineages. The lineage signs measure sensitivity to the frozen learned heads; they do not create 18 independent physical worlds. The physical replication count is six.
Verified fact	Cross-block P13	The Fable C2 contrast takes P13 from the H-A result block. The challenger verified matching world identity for all six seeds, including start epoch, initial-world identity, state, mask, and keyed field components.
Verified fact	No TEST or training	Both result files declare TRAIN worlds, <code>test_split_opened=false</code> , <code>episode_training=false</code> , and <code>learned_q2_used=false</code> .
Verified fact	Existing-head immutability	The runner snapshots the frozen heads and checks that parameters, training flags, optimizer/update state, and gradients do not change across arms.
Inference	Claim supported	The screen supports statements of the form “under these six matched TRAIN worlds, these frozen Q1/Q3 lineages, and this particular oracle-Q2 formula, FULL had the recorded direction relative to a DROP arm.”
Inference	Claim not supported	It does not establish that a learned C2 will reproduce the oracle surface, that the direction will hold on held-out worlds, or that the final three-head method improves deployment EE.

2.3 Frozen approximations

Verified fact. The H-A oracle and the Fable lane’s OPS-3 reading use unit future fading and shadowing, derive a constant interference-plus-noise block through inversion of the current gamma/SINR block, and hold the non-focal context fixed while evaluating candidate focal segments. The assumptions were fixed before outcomes and applied symmetrically across arms. See `V08-C2-STAGE1B-ORACLE-SCREEN-CONTRACT-2026-09-02.md:9-35` and `V08-C2-STAGE1B-ADDENDUM-OPS3-VARIANT-2026-09-02.md:19-36`.

Inference. These approximations weaken correspondence to the native live decision problem because future geometry, interference, D2 eligibility, fading, and non-focal

reactions can alter candidate ordering. They do not invalidate the screen as an evaluation of the approximate oracle surfaces actually declared.

Verified fact. The non-focal context is reconstructed from committed prior state. At the first step of a fresh episode, there is no prior committed outcome, so the Fable runner uses an empty/zero background. The census found materially worse predictive correspondence for continuing users at step 0—approximately 12.05% median relative error, compared with about 0.335% for continuing users after step 0. See `fable-cleanroom-lane/census/census-report.md:166-204`.

Inference. The step-0 convention is a representativeness limitation, not a pairing defect: all arms in the matched world encounter the same convention. It is nevertheless important because a ten-step episode gives the warm-start artifact non-negligible weight.

2.4 Service gate and the literal shortfall

Verified fact. The safe decision mask and physical service are distinct mechanisms. A candidate may be legal at decision time but fail the later per-link power-feasibility service gate. This is explicit in `source/env/service.py:40-50` and in the action-to-service path at `source/env/step.py:720-855`.

Verified fact. In H-A, FULL serves exactly one fewer user-step than DROP-C2 across 18,000 possible paired user-steps. That difference is not rounding: the raw counts are 17,958 versus 17,959. H-A also trails DROP-C3 by three served user-steps and ties DROP-C1. In the Fable OPS-3 block, FULL leads DROP-C2 by one but trails DROP-C3 by two and DROP-C1 by four. See `fable-cleanroom-lane/oracle/stage1b-report.md:181-203`.

Inference. One missed service step does not demonstrate a broad safety collapse, but it is an exact failure of the preregistered non-inferiority guard. It may not be dismissed because it is numerically small after observing the result.

2.5 Leakage audit

Label	Leakage type	Finding
Verified fact	Seed/world leakage	The keyed physical field root is a function of component and evaluation seed; it excludes arm, lineage, and policy label. Seeds and arm definitions were in frozen contracts before outcomes.
Verified fact	Target/outcome leakage	Q2 oracle values are constructed from predecision state and deterministic projections. The runner does not choose seeds, formulas, signs, or retained rows using final EE outcomes. Every target sign is retained.

Label	Leakage type	Finding
Verified fact	Old-Q2 leakage	The old learned Q2 is not used to score the H-A or Fable OPS-3 oracle arms. The runner explicitly ignores or zeros it.
Verified fact	Action leakage	Each arm receives a fresh environment instance and executes only the action selected by that arm's own direct summed surface. No other arm's selected action is injected into it.
Verified fact	Policy leakage	The Main reference computation is diagnostic or an action-independent centering reference; it is not a second executed policy and does not enter the Fable oracle action score.
Verified fact	Outcome mutation	The runner checks that frozen networks and associated state remain unchanged before and after episodes.
Inference	Residual RNG assurance limitation	The Fable runner passes the live environment RNG to a <code>main_actions(...)</code> diagnostic call, while the underlying production implementation of that call is not included in the review snapshot. The diagnostic result is not used in scoring or execution, and keyed fading after reset does not depend on that RNG, so there is no observed outcome leakage. Full exclusion of harmless RNG consumption cannot be independently reconstructed from this reduced package.
Inference	Overall leakage verdict	I found no target, seed, action, policy, sign, or outcome leakage that invalidates either completed direction screen. The remaining issues limit reproducibility and scope rather than falsifying the captured contrasts.

3. C2 candidate adjudication and ranking

3.1 Present-evidence ranking

Inference. Applying the required ordering—three-marginal direction and stability first, then held-out/null evidence, native coverage, simplicity, and compute—produces **no promotable C2 winner**. Both completed candidates fail the first method-level criterion because C3 is negative and the route-level service guard fails; the exact and option-value candidates lack outcome evidence.

Rank	Label	C2 object	Adjudication
1	Verified fact	H-A deterministic hold-horizon	It was run and gives the strongest completed C2 direction: +11.699953%, positive in 3/3 lineages and worlds. Its C3 marginal is the less-negative comple

Rank	Label	C2 object	Adjudication
		segment-timing surplus	value, -0.484836% , but still 0/3 lineages. Its C2 se contrast fails by one user-step. It is the simplest ar faster, approximately 8.27 seconds per episode.
2	Verified fact	Fable lane's preregistered OPS-3 reading	It was run and gives $+9.479997\%$ C2, positive in 3/ lineages and 6/6 worlds. C3 is more negative at -2.687311% , 0/3 lineages and 0/6 worlds. Overall : fails. It uses several non-native approximations and approximately 12.65 seconds per episode.
3	Verified fact	Current exact OPS-3 formula/live adapter	It has the best native coverage and reported mech evidence, including native D2 projection, cell-centr support, absorbing persistence, projected frozen background, exact Main centering, and fail-closed validation. Its declared 12- and 36-episode outcom worlds are unopened, so direction and stability are unknown.
4	Verified fact	Fable orbital option-value proposal	It remains unrun. Its support set, branch-local maximization, background equality, causal semant observability, and scale remain open. It cannot out candidates with completed or mechanics-closed evidence.

Inference. H-A ranks first only as the best **completed diagnostic**, not as the selected final C2. The present exact OPS-3 route remains the forward implementation candidate because it was independently defined before outcomes and offers the strongest native correspondence, not because it beat the completed candidates.

Proposal. Do not combine the favorable H-A result with selected pieces of Fable OPS-3 or exact OPS-3. Such a hybrid would be post-hoc. Retain the exact route as one coherent candidate, with only the independently preserved $h=0$ formula decision resolved before opening outcomes.

3.2 Why the Fable $+9.479997\%$ is not an exact-OPS-3 result

Label	Fable lane versus current exact OPS-3	Classification and consequence
Verified fact	Direct decision-step satellite positions versus cloning 47 native 0.640-second D2 updates per decision interval	Not a gauge change. It can alter geometry, D2 state, legal persistence, actions, and outcomes.

Label	Fable lane versus current exact OPS-3	Classification and consequence
Verified fact	No projected D2 eligibility versus explicit projected D2 eligibility	Not a gauge change. It can include or exclude candidate continuation at future offsets.
Verified fact	Focal-user elevation support versus physical cell-centre visibility	Not a gauge change. The two support predicates can disagree and change actions.
Verified fact	Independent per-offset feasibility/recovery versus absorbing persistence after first failure	Not a gauge change. Fable may award a later term after an intermediate failure; exact OPS-3 cannot.
Verified fact	Gamma-inverted constant interference-plus-noise versus interference from a projected, frozen committed non-focal background	Not a gauge change. Candidate SINR, rate, and marginal ordering can change.
Verified fact	Decimal <code>30.08</code> versus the canonical native clock produced by <code>47 × 0.640</code>	Not a gauge change. The binary floating-point values differ by one ULP. The physical effect is probably negligible away from boundaries, but it can affect exact receipts, ties, or threshold cases.
Verified fact	Omitted Main-reference subtraction versus exact Main-reference centering	A pure action-independent gauge provided the same valid reference is subtracted from every legal candidate at that anchor . It changes reported values but not argmax or executed outcome.
Verified fact	Permissive <code>_last_outcome</code> reconstruction versus fail-closed committed-state validation	Not a gauge change. On a valid state the intended score may coincide; on stale, malformed, or incomplete state the Fable route may silently score while the exact adapter rejects the anchor.
Inference	Overall relation	These are different C2 estimands and runtime semantics. The Fable O-arm result is evidence about the Fable lane's declared OPS-3 reading only, not about <code>source/runtime/ee_axis_ops3_live.py</code> .

The exact-versus-Fable comparison is documented in `INTEGRATION-VERIFICATION.md:60-83`, `docs/MULTI-CATFISH-C2-OPS3-FORMULA-PROBE-CONTRACT-2026-09-02.md:30-157`, and the Fable addendum.

3.3 Pre-outcome h=0 formula ruling

Verified fact. The current exact formula initializes persistence from the current legal mask and starts projected physical feasibility at $h=1$. The mechanics checkpoint explicitly preserved the question of whether future persistence must also require successful opening service at $h=0$, and classifies that choice as a formula decision rather than an adapter repair. See `docs/MULTI-CATFISH-C2-OPS3-MECHANICS-CHECKPOINT-2026-09-02.md:105-121` and `source/runtime/ee_axis_ops3.py:630-674`.

Inference. Legality is insufficient to establish a physical segment. Because execution applies a distinct per-link power-feasibility gate, an action may be legal but deliver no opening service. Crediting $h \geq 1$ persistence to a segment that did not open at $h=0$ is causally inconsistent with a segment-persistence interpretation.

Proposal. Future exact OPS-3 persistence must be conditioned on successful opening service:

$o_0(a) = \mathbf{1}\{\text{candidate } a \text{ is legal at the anchor}\} \mathbf{1}\{\text{the } h=0 \text{ execution-equivalent s}$

$$\chi_h(a) = o_0(a) \prod_{r=1}^h \rho_r(a), \quad h \geq 1,$$

where each ρ_r retains the existing exact D2, cell-visibility, gain, and power-feasibility conditions.

Proposal. If $o_0(a) = 0$, every future term for that candidate must take the existing outage/non-persistence branch; it may not receive later continuation credit.

Inference. This decision can change action ordering and outcomes. It therefore requires an OPS-3 formula version increment, updated hashes, and mechanics assertions before any declared outcome is opened. It is not a tunable response to the Fable signs, and it does not authorize importing H-A or Fable projection semantics into exact OPS-3.

4. C1/C3 diagnosis

4.1 C1

Verified fact. C1 remains positive in both completed new-Q2 contexts: +0.644142% under H-A and +5.891632% under the Fable lane's OPS-3 reading, each positive in 3/3 lineages. Its H-A physical-world sign is less uniform, 4/6, so this remains development evidence rather than confirmed efficacy.

Verified fact. The frozen λ_0 was calibrated from an r1-greedy Main-network rollout using objective weights `(1, 0, 0)`, not from the `(0.5, 0.3, 0.2)` Main policy described in older prose. The same frozen value was nevertheless used consistently in every applicable C1 row and arm. It cannot create a matching, pairing, or arm-specific arithmetic discrepancy.

C3 is λ_0 -invariant. See fable-cleanroom-lane/audit/c1c3-health-audit.md:31-62,254-291.

Inference. Changing λ_0 now would be outcome-responsive retuning and would invalidate the sealed comparability chain. The provenance mismatch does not justify redesigning C1.

Proposal. Make no C1 design or numerical change. Correct the documentation to describe the actual r1-greedy, one-episode TRAIN calibration provenance; keep the frozen constant and existing C1 receipts intact.

4.2 Current C3

Verified fact. Frozen C3 has the following independent negative evidence:

- H-A FULL versus DROP-C3: -0.484836%, negative in 3/3 lineages.
- Fable OPS-3 FULL versus DROP-C3: -2.687311%, negative in 3/3 lineages.
- Fresh Q1+C3 versus Q1: -9.715688%, reproducing the sealed -10.285558% direction.
- The old positive C3 results occurred only in combinations containing the now-retired harmful old Q2.

Inference. The isolated Q1+C3 result rules out the simple explanation that C3 is harmful only because a new, overly dominant Q2 suppresses it. Its previous positive margins were likely context-dependent compensation for the old Q2 rather than an independently complementary contribution.

Proposal. Retire the current C3 target/source from the next integration candidate. Do not reverse its sign, select a different rung, change its scale, or search for a favorable seed after these outcomes.

4.3 Proposed replacement C3 estimand

Proposal. Replace instantaneous non-focal rate difference with a **Projected Non-Focal Persistence Externality**. It should measure the current and short-horizon delivered-bit burden that successfully serving focal user u through candidate action a imposes on every other user.

Let $\mathcal{B}_{t,h}^{-u}$ be the fail-closed, committed non-focal service background projected to offset h under the same native clock and deterministic channel convention as exact OPS-3. Let $F_{u,a,h}$ be the focal physical segment, present only if it successfully opens at $h=0$ and remains persistent through offset h . Define the action-attributable non-focal bit externality

$$e_{3,h}(a) = \Delta t \left[\sum_{v \neq u} \hat{R}_{v,h} \left(\mathcal{B}_{t,h}^{-u} \oplus F_{u,a,h} \right) - \sum_{v \neq u} \hat{R}_{v,h} \left(\mathcal{B}_{t,h}^{-u} \right) \right].$$

For $H_t > 0$, use

$$X_3(a) = e_{3,0}(a) + \frac{1}{H_t} \sum_{h=1}^{H_t} e_{3,h}(a);$$

at a terminal anchor, use $X_3(a) = e_{3,0}(a)$. Centre it on the exact Main reference action:

$$Z_3^*(a) = X_3(a) - X_3(a^M), \quad Q_3^*(a) = \frac{Z_3^*(a)}{\kappa}.$$

Proposal. The no-focal background in both branches must be identical. The only branch difference may be insertion of the focal candidate segment. This directly tests the physical externality of that action rather than allowing background reconstruction differences to masquerade as C3 signal.

Proposal. Retain every sign. Do not clip the normally negative burden, negate it after observing outcomes, or tune κ . Main-reference centering is only the common action-independent gauge.

Non-overlap

Label	Head	Physical quantity
Verified fact	C1	h=0 focal delivered bits minus λ_0 times the h=0 total-network energy difference.
Verified fact	Exact C2	Projected focal continuation bits, projected marginal network energy, and focal persistence/service risk.
Proposal	New C3	h=0 and projected future non-focal delivered-bit externality only . No focal rate, no energy term, no standalone handover penalty, and no duplicate outage term.

Inference. This separates the final EE decomposition coherently. C1 and C2 represent the focal numerator and relevant energy denominator terms; C3 represents the missing non-focal numerator effect of the focal action.

Action-aligned state

Proposal. A future C3 learner state, if later authorized, must be built from the same live predecision anchor and indexed by each legal physical (NORAD, cell) action. It should include:

- committed same-beam non-focal victim count and delivered-bit burden;
- committed cochannel-victim count, delivered-bit burden, serving-signal burden, and interference susceptibility;
- committed non-focal RF-power and association information needed by the frozen-background projection;

- candidate-specific $h=0$ opening feasibility and exact projected persistence descriptors;
- projected candidate-to-victim coupling at each bounded offset.

Proposal. The state and target source may not read the current realized joint action, a successor outcome, another arm's action, future policy decisions, final EE, or the target sign. Missing or inconsistent committed state must fail closed rather than be reconstructed permissively.

Catfish source

Proposal. Construct the C3 Catfish source from fresh TRAIN anchors selected outcome-blind by fixed physical strata, such as:

- zero versus nonzero same-beam victims;
- zero versus nonzero cochannel exposure;
- short versus long projected focal persistence;
- D2/visibility boundary versus interior anchors;
- low versus high legal-action count.

All legal candidate actions at retained anchors should be evaluated, all signs retained, and quotas fixed before any C3 target or endpoint is inspected. The source must not rank anchors by Z_3^* , action flips, or final EE.

Common units and final-EE mechanism

Proposal. $e_{3,h}$, X_3 , and Z_3^* are delivered bits. Division by the same frozen κ places Q3 on the existing common normalized bits-equivalent axis. No separate Q3 scaling is permitted.

Inference. A focal action can lower other users' delivered bits through same-beam bandwidth sharing, beam activation, RF-power interactions, and cochannel interference. Because final network EE sums bits from **all** users, selecting an action with a smaller persistent non-focal burden can increase the ratio-of-sums numerator without adding a duplicated denominator term. Improvement is physically possible but not guaranteed.

5. Recommended method-level direction

Label	Candidate decision	Ruling
Proposal	Proceed to a bounded C2 learner gate while current C3 is held	Reject now. It would test learnability of one component while the required three-head method already has a replicated negative C3 marginal and while exact C2 outcome direction remains unopened.

Label	Candidate decision	Ruling
Proposal	Accept a two-head C1+C2 method	Reject now. Q1+Q2 versus Q1 is positive in the fresh oracle diagnostics, but these are TRAIN-only oracle comparisons, not a learned or held-out two-head result. Accepting two heads would be a scientifically different method requiring its own independent preregistration.
Proposal	Revise the acceptance structure	Reject now. Weakening the positive FULL-versus-each-DROP requirement after observing negative C3 outcomes would be post-hoc unless supported by a separately motivated estimand and a fresh evidence program.
Proposal	Stop C2 structurally	Reject. Two distinct C2 mechanisms are positive in every tested world and lineage. The evidence supports unresolved formula selection and learnability risk, not structural impossibility.
Proposal	Hold C2 and redesign C3	Select. It addresses the currently replicated method-level blocker while preserving the possibility of a genuine three-Catfish architecture.

Proposal. Preserve one common safe mask, direct unweighted $Q1 + Q2 + Q3$, one argmax, one executed Main action, canonical ratio-of-sums EE, and the service guard.

Proposal. Keep current exact OPS-3 provisional, with the h=0 opening-service condition incorporated as a versioned pre-outcome formula change. Do not open it as a standalone old-C3 experiment and do not train it yet.

6. Single preregistered next step

One integrated no-training oracle smoke

Proposal. Execute exactly one superseding gate: a **12-episode, one-unopened-TRAIN-world, three-lineage integrated oracle smoke** combining exact OPS-3 v1.1 with the proposed non-focal persistence C3. The executable should first run its frozen mechanics/receipt checks and, only if they all pass, run the paired rollouts. This is one gate, not separate adaptive experiments.

Candidate and formulas held fixed

Proposal.

- Q1: the existing frozen lineage-specific Q1 checkpoints, unchanged.
- Q2: current exact OPS-3 with only the pre-outcome h=0 opening-service condition added; native D2 projection, cell-centre visibility, absorbing persistence, exact clock,

frozen-background projection, horizon, outage term, λ_0 , and κ otherwise unchanged.

- Q3: the Projected Non-Focal Persistence Externality defined above, used as an oracle surface rather than a trained network.
- Old learned Q2 and current frozen Q3: prohibited from scoring any arm.

Arms and comparators

Proposal. For each lineage ℓ , run exactly these four arms:

$$\begin{aligned}\text{FULL}_\ell^* &= Q_{1,\ell} + O_2^{\text{exact}} + O_3^{\text{PNFE}}, \\ \text{DROP-C2}_\ell &= Q_{1,\ell} + O_3^{\text{PNFE}}, \\ \text{DROP-C3}_\ell &= Q_{1,\ell} + O_2^{\text{exact}}, \\ \text{DROP-C1}_\ell &= O_2^{\text{exact}} + O_3^{\text{PNFE}}.\end{aligned}$$

Proposal. Use one identical Boolean mask, direct left-to-right unweighted summation, one masked argmax, and one executed action. Main-reference actions may be used only for the fixed action-independent target gauge; no Main counterfactual is executed.

Proposal. Do not add pair-versus-singleton promotion conditions. The only binding EE comparisons are FULL versus the three DROP arms.

Unopened TRAIN world and lineage pairing

Proposal. Retain the already declared unopened exact-smoke seed `2026104501`; do not choose a new seed after seeing the Fable results.

Proposal. Pair the same three frozen lineages `2026092101`, `2026092102`, and `2026092103` on that physical world. Use fresh environment construction per arm but an identical start epoch, initial physical state, candidate table, mask, mobility stream, and keyed channel-field root. The keyed physical root must exclude arm, lineage, and policy labels.

Proposal. Run one ten-step episode for each of four arms and three lineages: 12 episodes total. DROP-C1 may be policy-identical across lineages because its two surfaces are oracle surfaces, but it should still be emitted in the uniform lineage receipt so every contrast has an explicit paired row.

Raw decision rules

Proposal. For each arm, pool raw delivered bits and raw network energy across the three lineage rows and calculate exactly once:

$$EE_{\text{arm}} = \frac{\sum_{\ell,t,u} B_{\ell,t,u}}{\sum_{\ell,t} E_{\ell,t}}.$$

Report pooled and per-lineage values; do not average percentages or episode EE ratios.

Proposal. The smoke passes the EE direction check only if all three pooled margins are strictly positive:

$$EE(\text{FULL}^*) > EE(\text{DROP} - C_j), \quad j = 1, 2, 3,$$

and each margin is positive in at least two of the three lineages.

Proposal. The service guard passes only if FULL has no lower pooled served-user-step count than each corresponding DROP arm and is non-inferior in at least two of three lineage pairs for every comparison. There is no numerical tolerance: a one-user-step shortfall fails.

Proposal. Record raw bits, raw energy, served-user-step counts, selected-action trace hashes, candidate-mask hashes, per-head surface hashes, action-flip counts, and all lineage-specific margins.

Hard stops

Proposal. Invalidate the gate immediately for any contract/hash mismatch, source-identity failure, non-common world, mask mismatch, nonfinite surface, reference action not exactly zero after centering, inconsistent frozen background, stale committed state, hidden mutation, TEST access, learner update, old-head use, future-policy query, or action crossing.

Proposal. Stop this C3 candidate without seed expansion if:

- any FULL-versus-DROP EE rule fails;
- any service rule fails;
- O_3^{PNFE} has zero legal-action spread throughout the world;
- FULL and DROP-C3 never select different actions, making the C3 marginal unexposed;
- h=0 service feasibility in the oracle disagrees with the execution-equivalent service predicate.

Proposal. A failure may not trigger a sign change, Q3 scaling change, horizon change, seed replacement, threshold relaxation, alternative rung, or fallback hybrid. A pass does not authorize a learner; it establishes only that the redesigned integrated oracle candidate survived one bounded mechanics-and-direction smoke.

Items frozen before any outcome

Proposal. The preregistration must contain:

1. exact Q2 v1.1 equations, including the h=0 opening-service indicator;
2. exact C3 equations, no-focal background, Main centering, persistence, terminal convention, and non-overlap statement;
3. λ_0 , κ , interval clock, horizon, outage semantics, and floating-point representations;

4. fail-closed committed-state and projected-background construction;
5. seed, start epoch, lineages, arm order, keyed-field namespace, and freshness rules;
6. raw ratio-of-sums and service decision rules;
7. action-exposure and hard-stop rules;
8. source, contract, runner, and result-schema hashes;
9. mechanics assertions for reference zero, candidate-equals-reference zero, $h=0$ execution agreement, common background, all-sign retention, no mutation, and no future-policy access.

Compute classification

Verified fact. The completed H-A and Fable screens required approximately 8.27 and 12.65 seconds per episode respectively, but neither computed a complete projected non-focal victim surface for every legal focal action.

Inference. Exact all-victim C3 projection adds a substantially larger action-by-user-by-horizon calculation, potentially including pairwise interference evaluation. A conservative estimate for the 12-episode gate is approximately **45–120 minutes** after implementation and vectorization.

Proposal. Classify this as **heavy** under the stated 30-minute boundary and run it on the Ubuntu server. It remains a no-training oracle rollout, not a learner gate.

7. Remaining risks and claim ceiling

Verified fact. Even a successful next smoke would still use one unopened TRAIN physical world, oracle Q2/Q3 surfaces, deterministic projected channel assumptions, and frozen non-focal behavior. It would not be a held-out result or learned-policy result.

Inference. The largest remaining risks would be:

- oracle-to-learner approximation error for both C2 and redesigned C3;
- inability of a compact causal state to predict the all-victim C3 surface;
- direct-sum scale imbalance—the census reports median Q2 action spread roughly 6.92 times the Q1+Q3 spread in the completed H-A setting;
- mismatch between a local additive bits-equivalent surrogate and the nonlinear episode-level ratio-of-sums endpoint;
- inaccuracies from unit fading, frozen background, short projection horizon, and episode step-0 initialization;
- service loss despite positive EE;
- incomplete source closure and inability to reproduce omitted original artifacts from this reduced package.

Inference. The Fable orbital option-value candidate remains a research proposal, not a fallback selected by these outcomes. Its native support, background equality, branch-max causal meaning, learnability, and interaction with C3 would require independent pre-outcome resolution.

Proposal. Treat the Fable lane's five shared-authority changes only as unapplied proposals:

1. add a post-Stage-1b authority notice describing C2 as direction-validated but blocked by the C3 method gate, and C3 as context-dependent;
2. correct the λ_0 sentence to state the actual `(1,0,0)` r1-greedy TRAIN calibration without recomputing it;
3. add the sealed `(Q1+Q3)-Q1 = -10.286%` conditional margin to a new versioned route-interaction diagnostic;
4. mark the old link-budget docstring claiming a non-binding feasibility gate as stale;
5. correct the presentation/figure handoff so the retired motion-one mechanism is not drawn as current C2.

Verified fact. None of those shared-authority patches was applied by the package. The copied `docs/CURRENT-MULTI-CATFISH-AUTHORITY.md` remains an earlier immutable orientation snapshot and cannot be cited as proof that V0.8 receipts were adopted.

Proposal. The claim ceiling after the proposed smoke, even if every rule passes, is:

On one preregistered matched TRAIN world and three frozen Q1 lineages, the exact $h=0$ -conditioned C2 oracle and the proposed projected non-focal C3 oracle produced positive FULL-versus-each-DROP ratio-of-sums directions without service inferiority.

It would still be impermissible to call that learned efficacy, held-out efficacy, deployment improvement, or a validated three-Catfish method.

8. Final decision

Inference. C2 has enough independent oracle-direction evidence to remain scientifically live, but not enough evidence for a learner gate. Current C3 has sufficiently replicated negative evidence to block the preferred three-head method independently of the old harmful Q2.

Proposal. Preserve C1, hold exact C2 at the oracle/formula level, replace the current C3 with a physically complementary projected non-focal externality, and make the single integrated no-training smoke above the next executable gate.

REDESIGN_C3_FIRST

